# Lazy ring + one directed shortcut: $\beta$ sweep, tail mechanism, and jump-over extension report

## Key conclusions and key figures

With $N = 100, q = 2/3, \rho = 1$ and only $\beta$ varied, both peaks move to earlier times and the tail rate $\gamma$ increases with $\beta$; at the same time, $K = 4$ sustains a wider bimodal window and a deeper valley. Figure 1 summarizes the peak and tail trends.

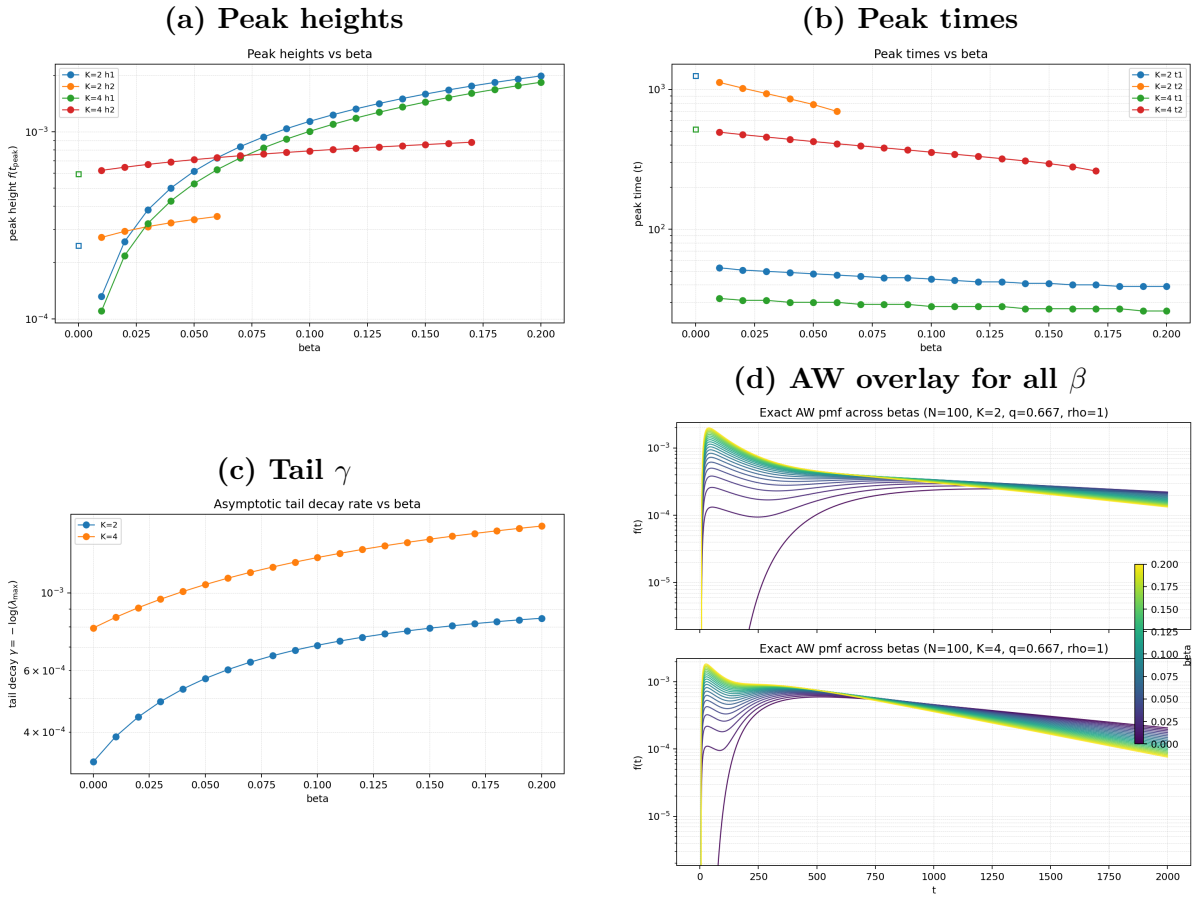


Figure 1: Exact AW results under a $\beta$-only sweep: peak heights, peak times, tail-decay rate $\gamma$, and the full-$\beta$ overlay of $f(t)$; $\beta = 0$ is the no-shortcut baseline and is shown as a standalone marker.

At $\beta = 0.01$, the representative $f(t)$ highlights the valley contrast: `hv_over_max` $\approx 0.34$ for $K = 2$ versus about 0.15 for $K = 4$, indicating a deeper valley and clearer separation for $K = 4$; at the same $\beta$, $K = 4$ has earlier $(t_1, t_2)$ and larger $\gamma$, consistent with faster time scales and lighter tails. Figure 2 shows the peak–valley window directly.

The tail mechanism is governed by the transient spectral radius, $\gamma = -\log(\lambda_{\max})$, and $\log S(t)$ is close to linear in the tail (Fig. 3).

MC classification confirms that jump-over appears only for $K = 4$. We sort trajectories by whether the shortcut is used at least once ($C0$ vs $C1p$) and whether a jump-over occurs ($J0$ vs

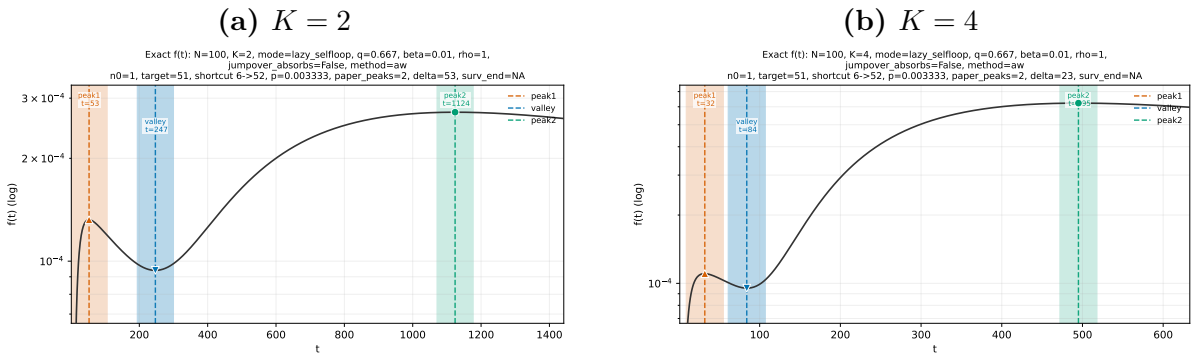


Figure 2: Representative exact $f(t)$ (AW), zoomed to the peak/valley window.

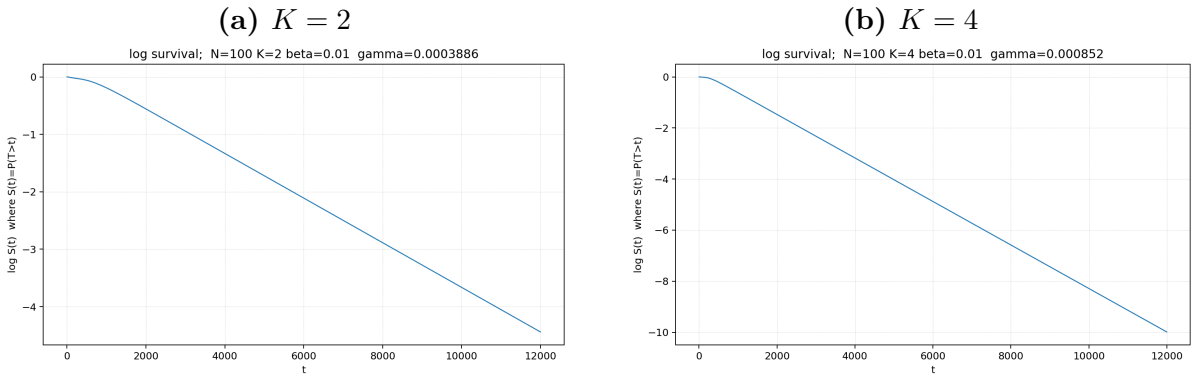


Figure 3: Log survival $\log S(t)$ at $\beta = 0.01$, consistent with the spectral-radius tail rate $\gamma$.

$J1p$), giving the four classes `C0J0`, `C1pJ0`, `C0J1p`, and `C1pJ1p`. Under this lens, $K = 2$ keeps $J$-related classes at zero across windows: peak1 is essentially all `C1pJ0`, the valley is mostly `C1pJ0` with a small `C0J0` tail, and peak2 is dominated by `C0J0` with a smaller `C1pJ0` share. For $K = 4$, peak1 splits into roughly two-thirds `C1pJ0` and one-third `C1pJ1p`; the valley shows a similar one-third $J1p$ share, while peak2 flips to `C0`-dominated with about a quarter `C0J1p`. These $J1p$ paths correspond to delayed crossings without immediate absorption, and the curves vary only mildly with $\beta$ (Fig. 4).

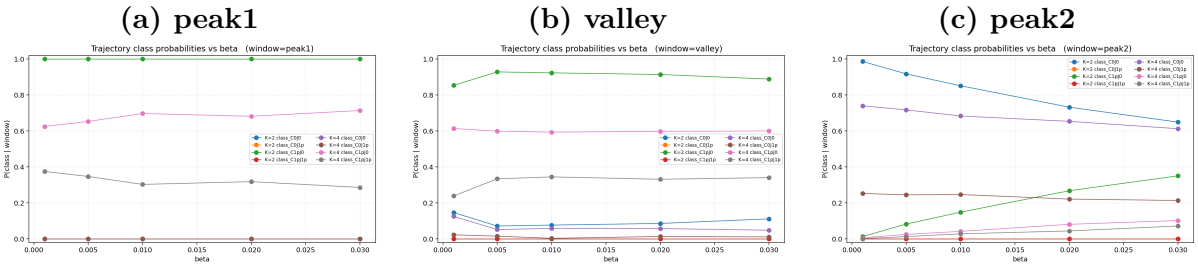


Figure 4: Conditional class probabilities vs $\beta$ for each window ($N = 100$).

To choose a representative bimodal setting, we apply $t_2/t_1 \geq 10$, `hv_over_max` $\leq 0.6$, and `h2_over_h1` $\geq 0.05$, selecting the smallest $\beta$ that satisfies both $K = 2$ and $K = 4$; the scan yields $\beta_* = 0.01$.

Fixing $\beta_* = 0.01$ and sweeping $N$ shows $t_2$ shifting right, $h_2$ decreasing, and $\gamma$ decreasing (heavier tails), while $t_1$ stays roughly stable; see Figures 5 and 6.

The MC sweep over $N$ sharpens the same contrast. For $K = 2$, everything remains in $J0$: peak1 is locked to `C1pJ0`, the valley is mostly `C1pJ0`, and peak2 mixes `C0J0` with `C1pJ0` as $N$ changes. For $K = 4$, once the valley/second peak are well formed, peak1 and the valley hold a stable ∼one-third $J1p$ share (largely `C1pJ1p`), while peak2 stays `C0`-dominated with about a quarter `C0J1p`, showing that jump-over contributions persist across $N$ (Fig. 7).

This extension builds on the main jump-over report (lazy_jump) and uses the same model

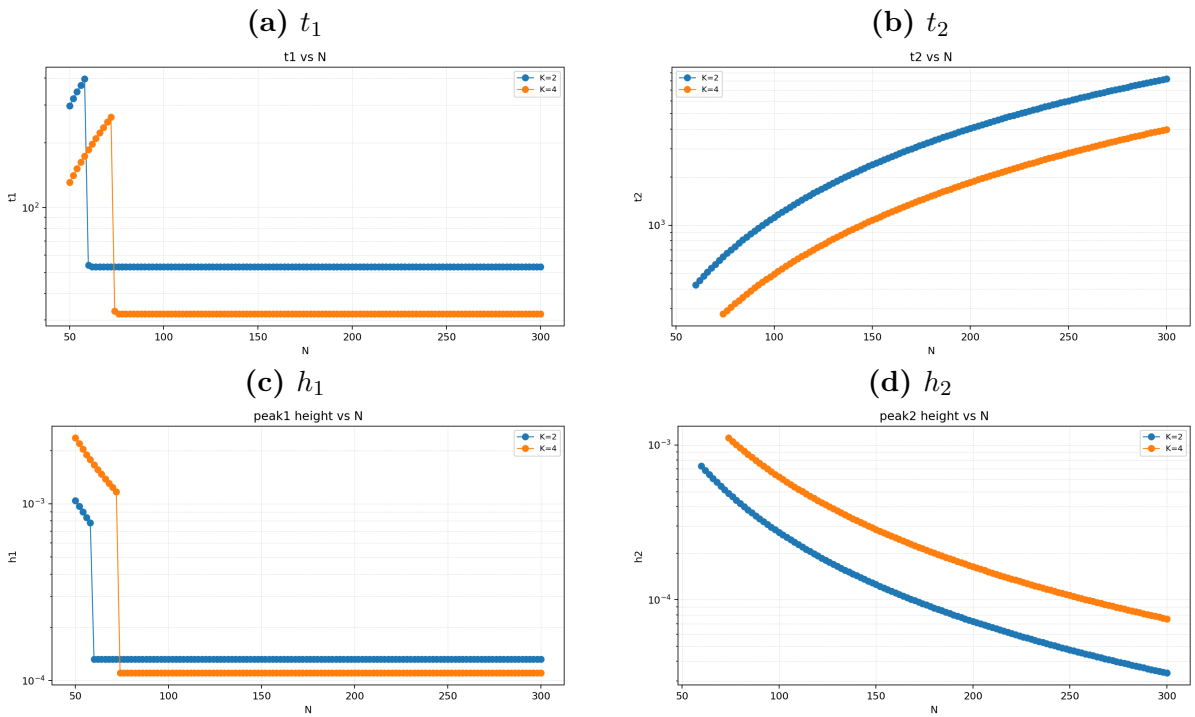


Figure 5: Peak locations and heights vs $N$ at $\beta = 0.01$.

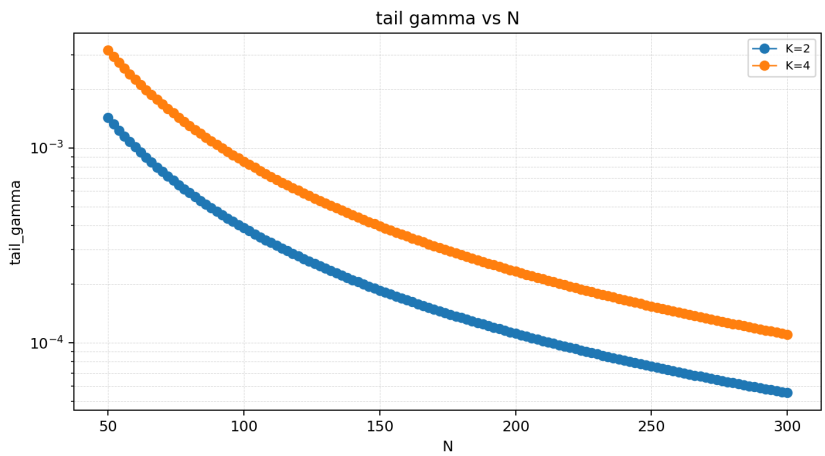


Figure 6: Tail decay rate $\gamma$ vs $N$ at $\beta = 0.01$.

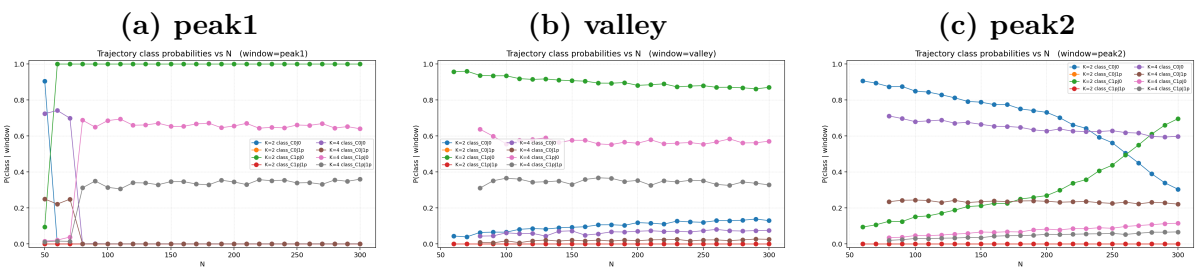


Figure 7: Class proportions vs $N$ at fixed $\beta = 0.01$.

and notation. The remainder gives the theory and valley definition, then the $\beta$ sweep, MC classes, $N$ sweep, MC-$N$, and reproducibility.

# 1 Theory: generating functions, AW inversion, and tail decay

This section provides the theoretical backbone for our exact pmf computation and tail diagnostics. We consider a lazy random walk on an $N$-node ring with $K = 2k$ neighbors: with probability $q$ the walker moves on the ring, choosing uniformly from steps $\pm 1, \ldots, \pm k$, and with probability $1 - q$ it stays put. A directed shortcut from src to dst is added.

## 1.1 Spectral decomposition and the resolvent without a defect

Let $P_0$ denote the defect-free transition matrix. For non-source nodes, the one-step transition is

$$P_0(i \to i) = 1 - q, \qquad P_0(i \to i \pm r) = \frac{q}{K}, \quad r = 1, \ldots, k.$$

Because $P_0$ is translation-invariant (a circulant matrix), its eigenvectors are Fourier modes

$$\phi_\ell(x) = \exp\left(\frac{2\pi i \ell x}{N}\right), \qquad \ell = 0, \ldots, N-1,$$

with eigenvalues

$$\lambda_\ell = (1-q) + \frac{q}{K}\sum_{r=1}^{k}\left(e^{2\pi i \ell r/N} + e^{-2\pi i \ell r/N}\right) = (1-q) + \frac{2q}{K}\sum_{r=1}^{k}\cos\left(\frac{2\pi \ell r}{N}\right). \tag{1}$$

Define the resolvent (matrix generating function)

$$S^{(0)}(z) = (I - zP_0)^{-1}, \qquad |z| < 1.$$

By spectral decomposition,

$$S^{(0)}(z)_{x,y} = \frac{1}{N}\sum_{\ell=0}^{N-1}\frac{\exp\left(\frac{2\pi i \ell (x-y)}{N}\right)}{1 - z\lambda_\ell}. \tag{2}$$

One may further define a displacement form $Q(d, z) := S^{(0)}(z)_{y+d,y}$ to efficiently assemble defect quantities.

## 1.2 Directed shortcut under the self-loop rule: a rank-one update

In the `lazy_selfloop` mode, the shortcut probability is

$$p = \beta(1-q),$$

and the same amount is removed from the self-loop probability at the source: $P(\text{src} \to \text{src}) = (1-q) - p$ and $P(\text{src} \to \text{dst}) = p$. Hence the transition matrix is a rank-one perturbation of $P_0$:

$$P = P_0 + p\, e_{\text{src}}(e_{\text{dst}} - e_{\text{src}})^\top, \tag{3}$$

where $e_i$ is the standard basis vector. Therefore

$$I - zP = (I - zP_0) - zp\, e_{\text{src}}(e_{\text{dst}} - e_{\text{src}})^\top.$$

Applying the Sherman–Morrison formula

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u},$$

one can obtain the required entries of the defect resolvent $S(z)$ from $S^{(0)}(z)$ with $O(1)$ additional scalar operations (as implemented in the repository pipeline).

### 1.3 First absorption generating function under geometric absorption ($\rho$)

Let tar be the target. Upon each visit to tar, absorption occurs with probability $\rho$, otherwise the process continues with probability $1-\rho$. Let

$$S_{i,j}(z) = \sum_{t\geq 0} \mathbb{P}(X_t = i \mid X_0 = j) z^t$$

be the visit generating function of the non-absorbed walk. Then the generating function of the first absorption pmf $f(t) = \mathbb{P}(T = t)$ for $t \geq 1$ is

$$\tilde{f}(z) = \sum_{t\geq 1} f(t) z^t = \frac{\rho\, S_{\mathrm{tar},n_0}(z)}{(1-\rho) + \rho\, S_{\mathrm{tar,tar}}(z)}. \tag{4}$$

For $\rho = 1$, this reduces to

$$\tilde{f}(z) = \frac{S_{\mathrm{tar},n_0}(z)}{S_{\mathrm{tar,tar}}(z)}.$$

### 1.4 Abate–Whitt (AW) inversion via FFT

Coefficient extraction can be discretized on a circle. With $z_m = re^{2\pi i m/L}$ $(r < 1)$,

$$f(t) \approx r^{-t} \frac{1}{L} \sum_{m=0}^{L-1} \tilde{f}(z_m) e^{-2\pi i m t/L}. \tag{5}$$

The right-hand side is an inverse DFT, thus FFT yields all $f(t)$ for $t = 1, \ldots, T_{\max}$ in $O(L \log L)$ time. We follow the repository's AW parameter selection and numerical stabilization.

### 1.5 Exponential tail decay and the spectral-radius rate $\gamma$

Let $Q$ be the transient submatrix of the absorption chain. The survival function is

$$S(t) = \mathbb{P}(T > t) = \mathbf{1}^\top Q^t e_{n_0}.$$

By Perron–Frobenius theory for nonnegative matrices, there exists $\lambda_{\max} \in (0,1)$ such that

$$S(t) \sim c\, \lambda_{\max}^t \quad (t \to \infty).$$

We define the asymptotic exponential tail decay rate

$$\gamma = -\log(\lambda_{\max}), \tag{6}$$

which provides a stable and comparable tail metric across $\beta$, $N$, and $K$. Numerically, we construct $Q$ and compute its spectral radius; for $\rho = 1$ we remove the target state, and for $\rho < 1$ we multiply the target row by $(1-\rho)$ to represent absorption leakage.

## 2 Comparable definition of the valley and windowing

Let $f(t) = \mathbb{P}(T = t)$ for $t \in \mathbb{N}^+$ denote the pmf of the first absorption time. We detect candidate peaks as strict local maxima:

$$f(t) > f(t-1), \qquad f(t) > f(t+1), \qquad f(t) \geq h_{\min}. \tag{7}$$

We further filter secondary peaks by a relative-height threshold: $f(t) \geq$ `second_rel_height` $\cdot \max_u f(u)$.

### 2.1 V0: Default valley (minimum between the two peaks)

Given the first two peaks in chronological order $t_1 < t_2$, define

$$t_v = \arg\min_{t_1<t<t_2} f(t). \tag{8}$$

This definition is parameter-light and guarantees that $t_v$ lies strictly between the two peaks, facilitating comparisons across $\beta$ and across $K$.

### 2.2 V1: Relative-distance window scale (for trajectory-type comparisons)

Let $\Delta T = t_2 - t_1$ be the peak separation and define the half window width

$$\Delta = \max\left(1, \lfloor \delta_{\text{frac}} \Delta T \rfloor\right). \tag{9}$$

Each sample is assigned to the nearest center in $\{t_1, t_v, t_2\}$ if $\min_{c\in\{t_1,t_v,t_2\}} |T-c| \le \Delta$, otherwise it is assigned to `other`. Because $\Delta$ scales proportionally with $\Delta T$, the windowing remains comparable across $\beta$.

### 2.3 V2: Robust valley (excluding regions close to peaks)

To avoid valleys too close to a peak (which may happen when peaks are close), we may define

$$t_v = \arg\min_{t_1+\alpha\Delta T \le t \le t_2-\alpha\Delta T} f(t), \tag{10}$$

with $\alpha \in (0, 1/2)$ (e.g., $\alpha = 0.1$). We use V2 as a sensitivity-analysis convention.

## 3 Beta-only sweep at fixed $N$ ($N = 100$)

We fix $N = 100$, $q = 2/3$, $\rho = 1$, and the `lazy_selfloop` convention, and vary only $\beta$. Figure 1 summarizes exact AW peak heights/locations and the tail-decay metric, together with an all-$\beta$ overlay of $f(t)$.

**Peak–valley quantification** We record the valley depth `hv_over_max` $= h_v/\max(h_1, h_2)$ and `h2_over_h1` (see CSV). Comparing $K = 2$ and $K = 4$ highlights the depth difference in the valley. At $\beta = 0.01$, $K = 2$ has `hv_over_max` $\approx 0.34$, while $K = 4$ yields $\approx 0.15$, indicating a deeper valley and clearer separation for $K = 4$. At the same $\beta$, $K = 4$ shows earlier $t_1, t_2$ and larger $\gamma$, consistent with faster time scales and lighter tails. Note that $K = 2$ has $J \equiv 0$ (no jump-over), while $K = 4$ allows $J > 0$, which will reshape class compositions in the MC sweep below.

**Representative $f(t)$ with improved readability** Figure 2 shows $\beta = 0.01$ with the time axis focused on the peak–valley interval (avoiding the long vertical stretch near $t \approx 0$), making the peak/valley contrast visually clear.

**Tail decay diagnostics** The asymptotic tail is controlled by $\gamma = -\log(\lambda_{\max})$. Figure 3 confirms the near-linear decay of $\log S(t)$. At $\beta = 0.01$, $K = 2$ has $\gamma \approx 3.89 \times 10^{-4}$ while $K = 4$ has $\gamma \approx 8.52 \times 10^{-4}$, matching the $\gamma$–vs–$\beta$ curves.

### 3.1 Automatic choice of $\beta$

We apply the same thresholds ($t_2/t_1 \ge 10$, `hv_over_max` $\le 0.6$, `h2_over_h1` $\ge 0.05$) and choose the smallest $\beta$ that satisfies them for both $K = 2$ and $K = 4$. The scan yields $\beta_* = 0.01$ (see `outputs/selected_beta.txt`).

These exact AW results set the stage for the MC class analysis in the next section.

## 4 MC sweep: four classes vs $\beta$ ($N = 100$)

We use MC to estimate class probabilities within each window, defined by $(t_1, t_v, t_2)$ and $\Delta = \lfloor \delta_{\text{frac}}(t_2 - t_1) \rfloor$. The four mutually exclusive classes are: `class_C0J0` (no shortcut, no jump-over), `class_C1pJ0` (shortcut, no jump-over), `class_C0J1p` (no shortcut, jump-over), `class_C1pJ1p` (shortcut and jump-over). For $K = 2$ we have $J \equiv 0$, so $J$-related classes should be 0 or nearly 0. Figure 4 reports the conditional class probabilities across windows as $\beta$ varies.

**Interpretation highlights** For $K = 2$, $J$-related classes vanish, confirming that jump-over is absent. For $K = 4$, the valley and second-peak windows show larger shares of $J$-related classes, consistent with trajectories that cross the target without absorbing and hence contribute to delayed arrivals.

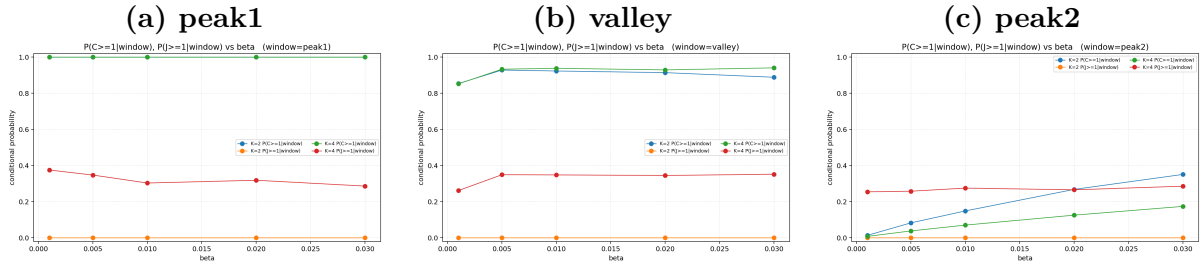


Figure 8: $\mathbb{P}(C \geq 1 \mid \text{window})$ and $\mathbb{P}(J \geq 1 \mid \text{window})$ vs $\beta$.

With the $\beta$-direction class structure established, we now fix $\beta_*$ and examine how the slow time scale varies with $N$.

## 5 Fix $\beta = 0.01$ and sweep $N$ (exact AW)

Using the automatically selected $\beta_* = 0.01$ and fixing $q = 2/3$, $\rho = 1$, and `lazy_selfloop`, we sweep even $N$ and track peak locations/heights and tail decay. Figure 5 shows that $t_2$ shifts right as $N$ grows (a slower second time scale), while $h_2$ decreases; the tail rate $\gamma$ decreases with $N$, indicating a heavier tail, and the trend is summarized in Figure 6.

The next section complements this with MC class statistics across $N$.

## 6 MC sweep: four classes vs $N$ (fixed $\beta = 0.01$)

With $\beta = 0.01$ fixed, we scan even $N$ and estimate class proportions within each window. Recall: $C$ records whether the shortcut is used at least once ($C0$ vs $C1p$) and $J$ records whether a jump-over occurs ($J0$ vs $J1p$), so the four classes are `C0J0`, `C1pJ0`, `C0J1p`, and `C1pJ1p`. Figure 7 summarizes the class proportions vs $N$.

**Comparison highlights** For $K = 2$, $J$-related classes are essentially zero; for $K = 4$, $J$-related classes appear prominently in the valley and second-peak windows, reflecting jump-over contributions to delayed absorption.

See the reproducibility section for the exact commands and parameters.

## 7 Reproducibility (commands and outputs)

Run the following inside `reports/lazy_jump_ext/`.

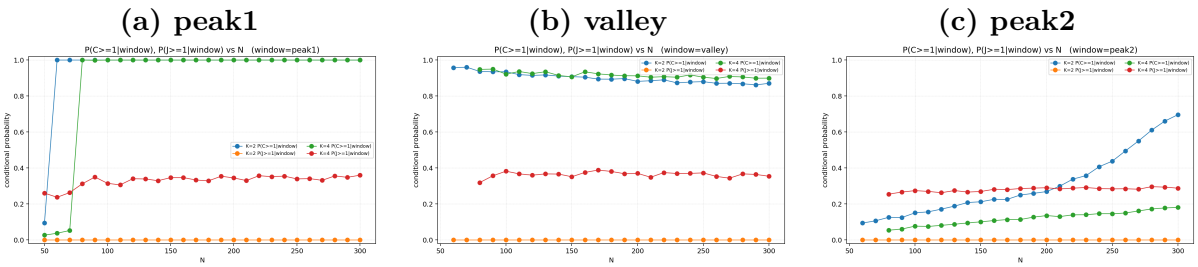


Figure 9: $\mathbb{P}(C \geq 1 \mid \text{window})$ and $\mathbb{P}(J \geq 1 \mid \text{window})$ vs $N$.

### 1. $\beta$ sweep (exact AW)

```
python code/beta_sweep_peaks_tail.py \
  --N 100 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --beta-min 0.0 --beta-max 0.2 --beta-num 21 \
  --outdir outputs/beta_sweep_N100
```

Outputs: `outputs/beta_sweep_N100/beta_sweep_metrics_N100.csv` and the three plots.

### 2. Automatic $\beta$ selection

```
python code/choose_beta.py \
  --csv outputs/beta_sweep_N100/beta_sweep_metrics_N100.csv \
  --Ks 2 4 --macro_ratio 10 --hv_over_max 0.6 --min_h2_over_h1 0.05 \
  --fallback_beta 0.02 \
  --out outputs/selected_beta.txt
```

### 3. Fix $\beta$ and sweep $N$ (exact AW)

```
python code/N_sweep_peaks_tail.py \
  --K 2 4 --beta 0.01 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --N-min 50 --N-max 300 --N-step 2 --only-even \
  --outdir outputs/N_sweep_beta_selected
```

### 4. MC sweep over $\beta$ (four classes)

```
python code/mc_sweep_beta_classes.py \
  --N 100 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --betas 0.001 0.005 0.01 0.02 0.03 \
  --n-walkers 200000 --seed 0 \
  --outdir outputs/mc_beta_sweep_N100
```

### 5. MC sweep over $N$ (fixed $\beta$)

```
python code/mc_sweep_N_classes.py \
  --beta 0.01 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --N-min 50 --N-max 300 --N-step 10 --only-even \
  --n-walkers 200000 --seed 0 \
  --outdir outputs/mc_N_sweep_beta_selected
```

**6. Tail diagnostics ($\log S(t)$)**

```
python code/tail_diagnostics.py \
  --N 100 --K 2 --beta 0.01 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --max-steps-aw 12000 --outdir outputs/tail_diag
python code/tail_diagnostics.py \
  --N 100 --K 4 --beta 0.01 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --max-steps-aw 12000 --outdir outputs/tail_diag
```

**7. Overlay all-$\beta$ exact curves (optional)**

```
python code/plot_aw_all_betas.py \
  --N 100 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --beta-min 0.0 --beta-max 0.2 --beta-num 21 \
  --out outputs/beta_sweep_N100/aw_all_betas_N100.png
```

**8. Build the extension PDFs**

```
latexmk -xelatex -interaction=nonstopmode ring_lazy_jump_ext_cn.tex
latexmk -pdf -interaction=nonstopmode ring_lazy_jump_ext_en.tex
```